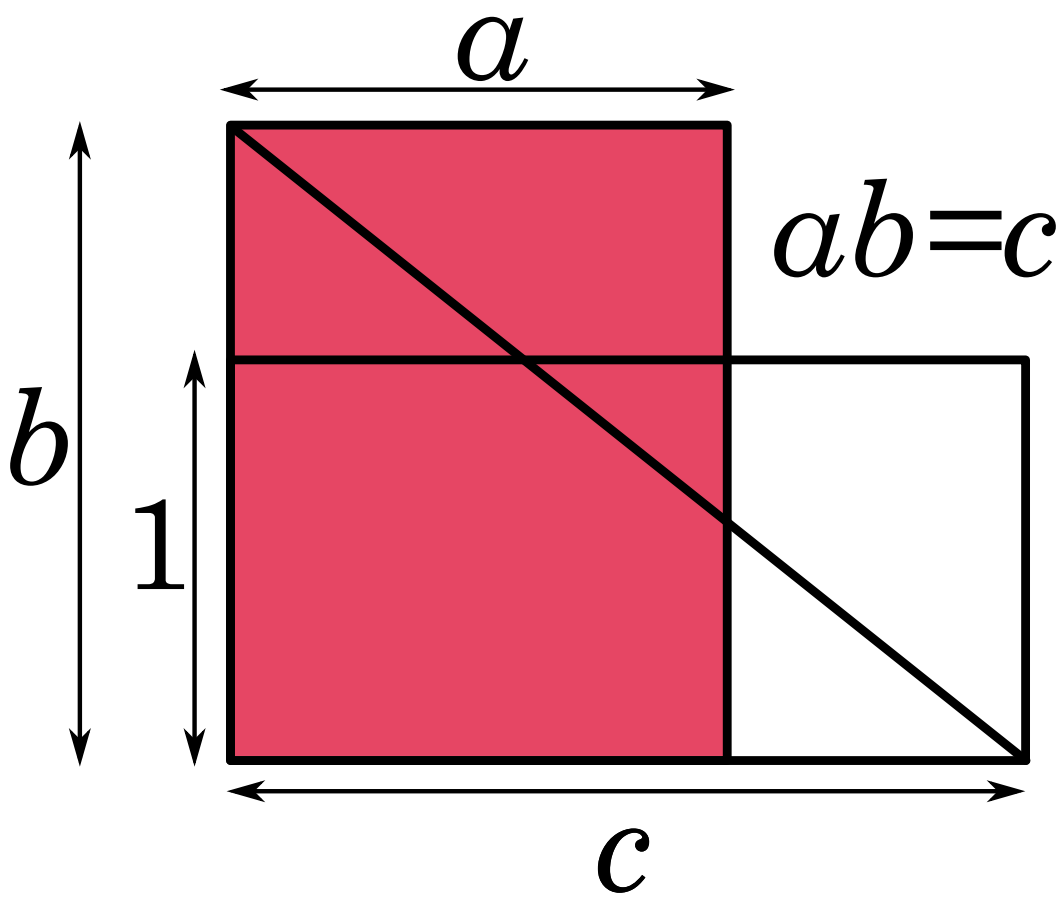
# Decomposing Polyhedra

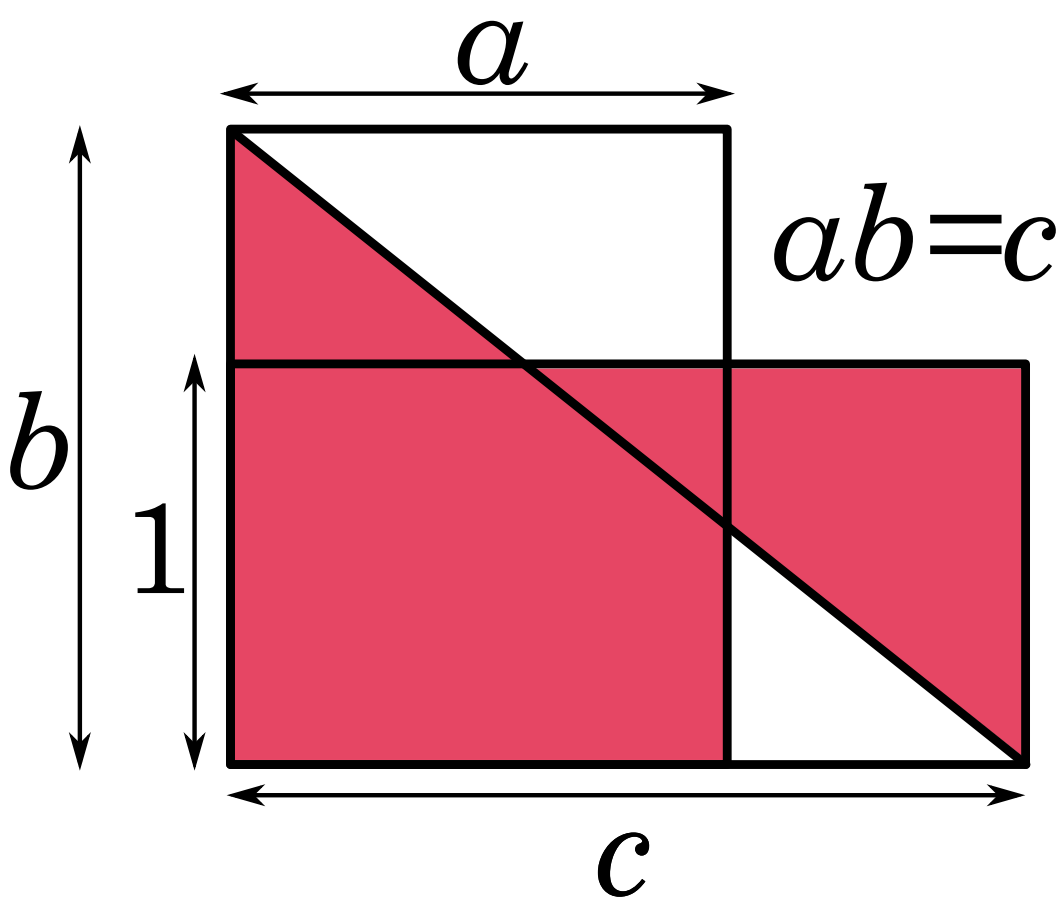
Undergraduate Mathematics Symposium
UIC ∘ October 1, 2011

Yvonne Lai
University of Michigan

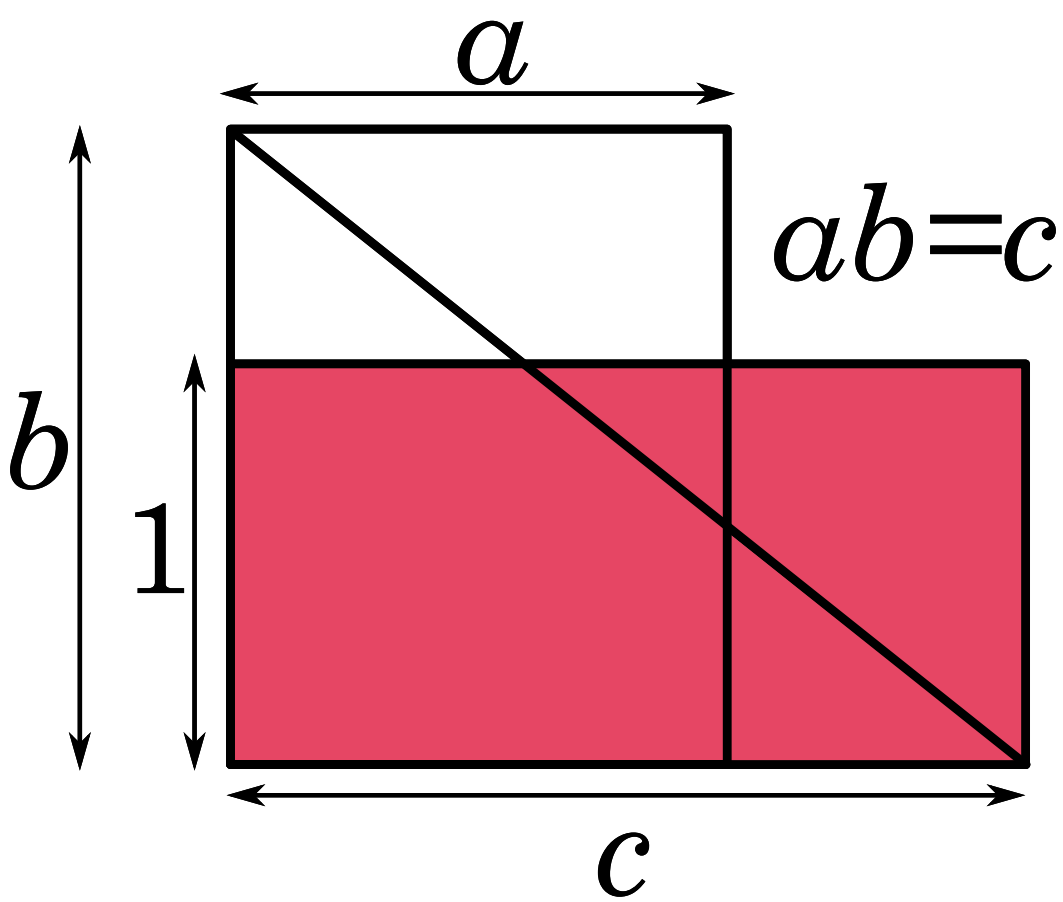
# 2-D Scissors Congruence



# 2-D Scissors Congruence



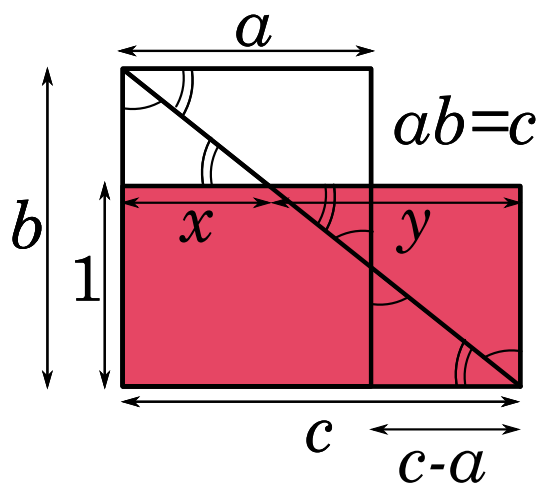
# 2-D Scissors Congruence



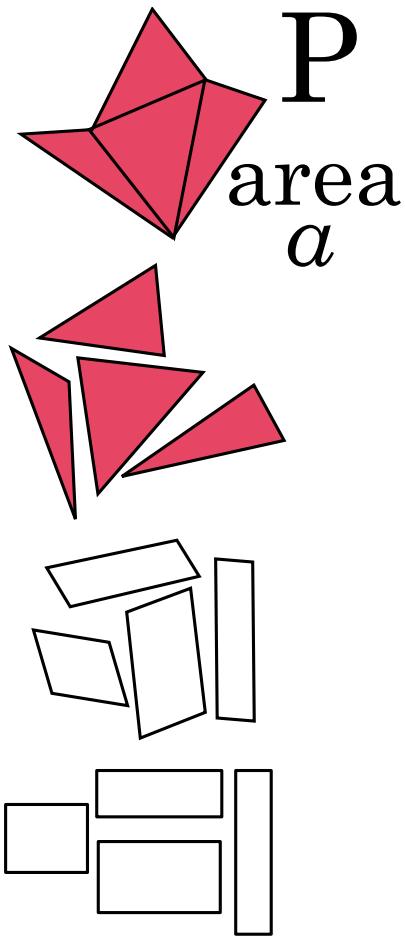
# 2-D Scissors Congruence

**Given $c = ab$, an $a \times b$ rectangle is *scissors congruent* to a $1 \times c$ rectangle.**

$$\begin{aligned}
\frac{x}{b-1} &= \frac{c}{b} \\
\Longrightarrow \quad x &= \frac{c(b-1)}{b} \\
&= \frac{ab(b-1)}{b} \\
&= ab - a \\
&= c - a.
\end{aligned}$$

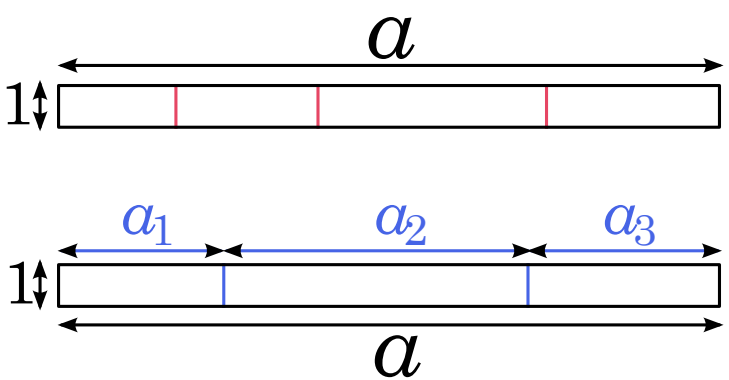


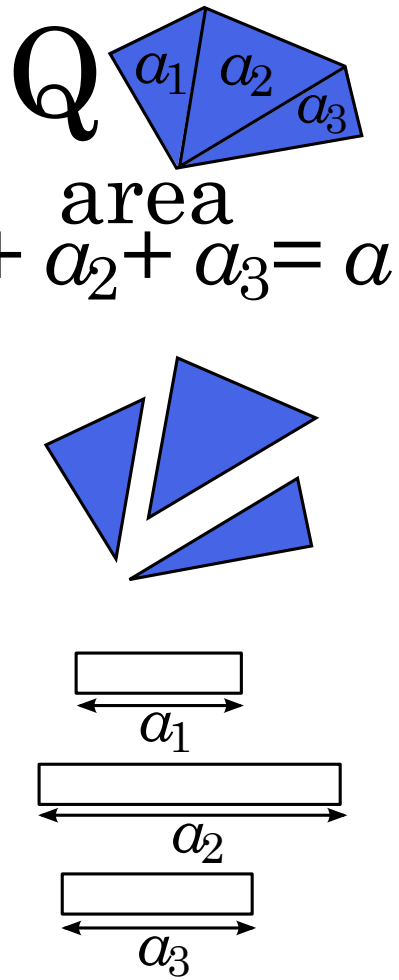
# 2-D Scissors Congruence




**Theorem.** If $P$, $Q$ are polygons of equal area, then we can cut $P$ into a finite number of smaller polygons, and rearrange the pieces to get $Q$.

(Bolyai-Gerwein 1833)



# 2-D case

**In 2 dimensions –**

- **cutting:** decomposing into a finite number of smaller polygons
- **rearranging:** moving with rigid motions (isometries)
- **scissors congruence ($P \sim Q$):** Two polygons $P$ and $Q$ are scissors congruent if you can cut $P$ and rearrange the pieces to form $Q$.
- **Theorem (2-D).** Let $P$ and $Q$ be polygons of equal area. Then $P \sim Q$. (Bolyai-Gerwien 1833)

What about 3 dimensions?

# 3-D Scissors Congruence

**In 3 dimensions –**

- **cutting:** decomposing into a finite number of smaller polyhedra
- **rearranging:** moving with rigid motions (isometries)
- **scissors congruence ($P \sim Q$):** Two polyhedra $P$ and $Q$ are scissors congruent if you can cut $P$ and rearrange the pieces to form $Q$.
- **??? Theorem (3-D).** Let $P$ and $Q$ be polyhedra of equal volume. Then $P \sim Q$. ???

# 3-D Scissors Congruence

**In 3 dimensions –**

- **cutting:** decomposing into a finite number of smaller polyhedra
- **rearranging:** moving with rigid motions (isometries)
- **scissors congruence ($P \sim Q$):** Two polyhedra $P$ and $Q$ are scissors congruent if you can cut $P$ and rearrange the pieces to form $Q$.
- ~~**??? Theorem (3-D).** Let $P$ and $Q$ be polyhedra of equal volume. Then $P \sim Q$. ???~~

# 3-D Scissors Congruence

**Polyhedron $P$**

- edges $E_1, E_2, \ldots, E_n$
- lengths $\ell(E_1), \ell(E_2), \ldots, \ell(E_n)$
- dihedral angles $\theta(E_1), \theta(E_2), \ldots, \theta(E_n)$.

**Keep track of information via formal sum of pairs:**

$$\ell(E_1) \otimes \theta(E_1) + \cdots + \ell(E_n) \otimes \theta(E_n)$$

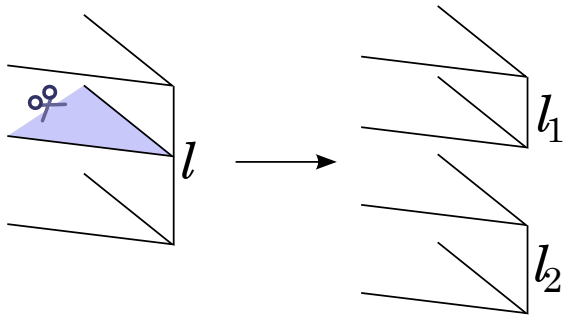
(Dehn Invariant for polyhedra).

# 3-D Scissors Congruence: Cutting and $+$, $\otimes$

**Property 1.**

$\ell = \ell_1 + \ell_2$

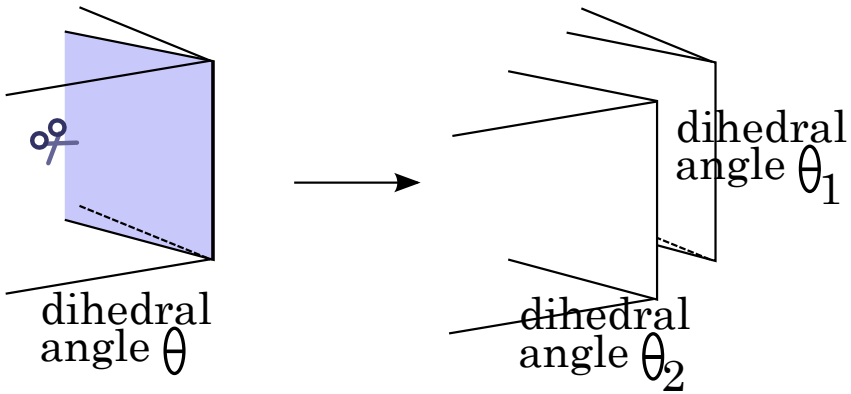
$\longrightarrow \ell \otimes \theta = \ell_1 \otimes \theta + \ell_2 \otimes \theta$




**Property 2.**

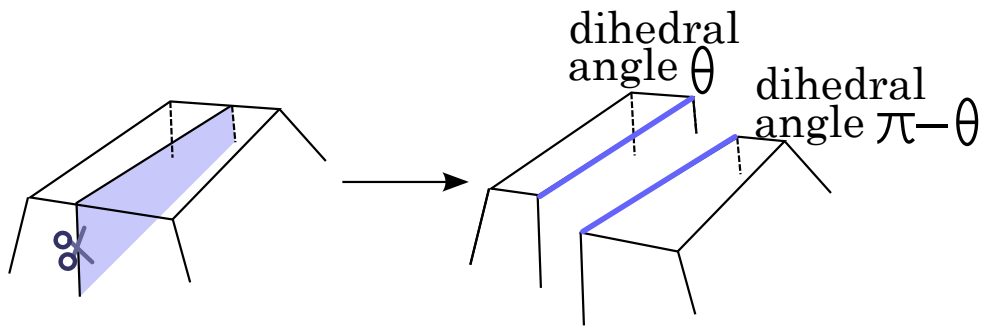
$\theta = \theta_1 + \theta_2$

$\longrightarrow \ell \otimes \theta = \ell \otimes \theta_1 + \ell \otimes \theta_2$




**Property 3.**

$0 = \ell \otimes \theta + \ell \otimes (\pi - \theta)$



# 3-D: Cutting and the Dehn Invariant

Polyhedron $P$ with Dehn invariant $\ell(E_1) \otimes \theta(E_1) + \cdots + \ell(E_n) \otimes \theta(E_n)$.

- **Cutting does not change the Dehn invariant.**
- $0 = \ell \otimes \pi$. Why:

$$0 = \ell \otimes \theta + \ell \otimes (\pi - \theta) = \ell \otimes \theta + \ell \otimes \pi - \ell \otimes \theta = \ell \otimes \pi.$$

- $0 = \ell \otimes r\pi$ for any rational $r$. (!) Why:

$$\text{Let } r = \frac{p}{q}. \implies \quad \ell \otimes \frac{p}{q}\pi$$

$$= \underbrace{\frac{\ell}{q} \otimes \frac{p}{q}\pi + \cdots + \frac{\ell}{q} \otimes \frac{p}{q}\pi}_{q \text{ times}}$$

$$= \frac{\ell}{q} \otimes (\frac{p}{q}\pi + \cdots + \frac{p}{q}\pi)$$

$$= \frac{\ell}{q} \otimes p\pi = \underbrace{\frac{\ell}{q} \otimes \pi + \cdots + \frac{\ell}{q} \otimes \pi}_{p \text{ times}} = 0$$

# 3-D: Cutting and the Dehn Invariant

Polyhedron $P$ with Dehn invariant $\ell(E_1) \otimes \theta(E_1) + \cdots + \ell(E_n) \otimes \theta(E_n)$.

- **Cutting does not change the Dehn invariant.**
- $0 = \ell \otimes \pi$. Why:

$$0 = \ell \otimes \theta + \ell \otimes (\pi - \theta) = \ell \otimes \theta + \ell \otimes \pi - \ell \otimes \theta = \ell \otimes \pi.$$

- $0 = \ell \otimes r\pi$ **for any rational** $r$. (!) Why:

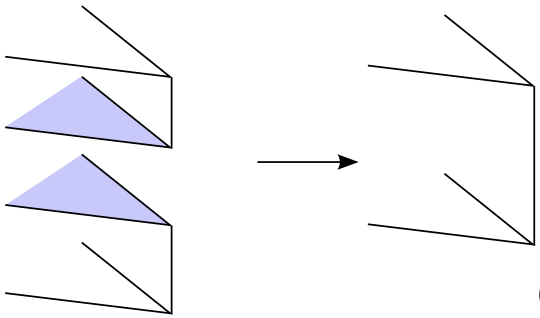
$$\text{Let } r = \frac{p}{q}. \implies \quad \ell \otimes \frac{p}{q}\pi$$

$$= \underbrace{\frac{\ell}{q} \otimes \frac{p}{q}\pi + \cdots + \frac{\ell}{q} \otimes \frac{p}{q}\pi}_{q \text{ times}}$$

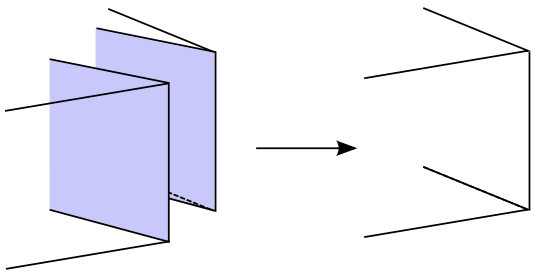
$$= \frac{\ell}{q} \otimes (\frac{p}{q}\pi + \cdots + \frac{p}{q}\pi)$$

$$= \frac{\ell}{q} \otimes p\pi = \underbrace{\frac{\ell}{q} \otimes \pi + \cdots + \frac{\ell}{q} \otimes \pi}_{p \text{ times}} = 0$$

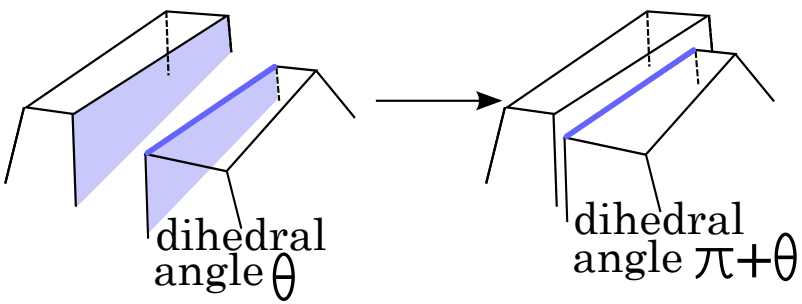
# 3-D Scissors Congruence: Gluing and $+$, $\otimes$



Check: $\ell_1 \otimes \theta + \ell_2 \otimes \theta = (\ell_1 + \ell_2) \otimes \theta$



Check: $\ell \otimes \theta_1 + \ell \otimes \theta_2 = \ell \otimes (\theta_1 + \theta_2)$




Check: $\ell \otimes \theta = \ell \otimes (\pi + \theta)$

**Gluing does not change the Dehn invariant.**

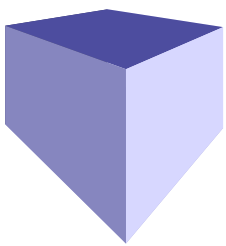
# 3-D Scissors Congruence

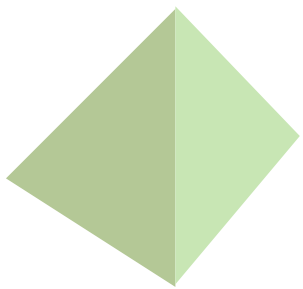
**In 3 dimensions –**

- **cutting:** decomposing into a finite number of smaller polyhedra
- **rearranging:** moving with rigid motions (isometries)
- **scissors congruence ($P \sim Q$):** Two polyhedra $P$ and $Q$ are scissors congruent if you can cut $P$ and rearrange the pieces to form $Q$.

**Theorem (3-D Scissors Congruence).** Let $P$ and $Q$ be polyhedra. If $P$, $Q$ have different Dehn Invariants, they are NOT scissors congruent ($P \not\sim Q$).

(Dehn 1900, in response to Hilbert.)



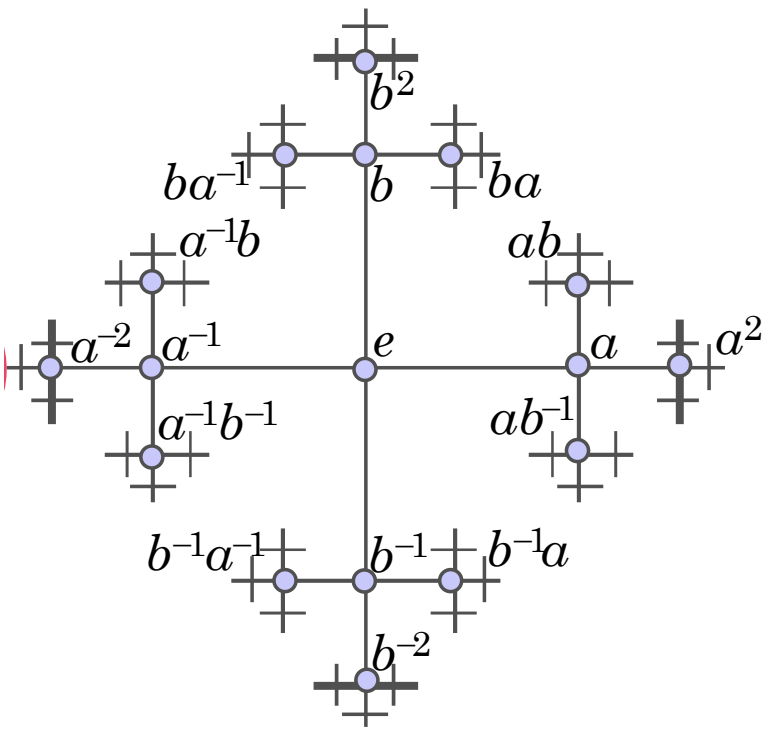


(Homework: Show that $\arccos \frac{1}{3}$ is irrational.)

What about more generalized decompositions?

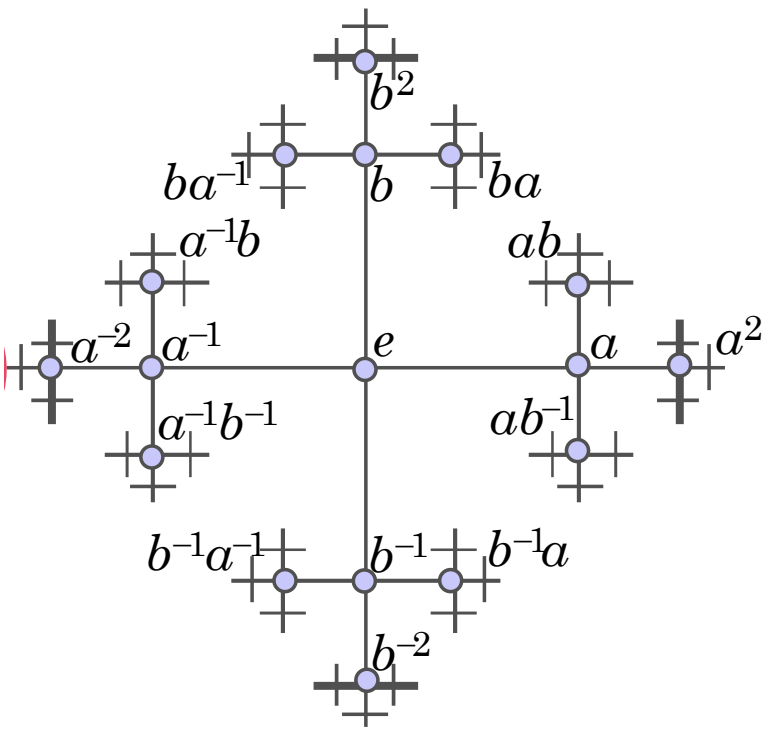
# Generalizing Scissors Congruence: Graphs and Alphabets

- **Words:** Finite strings of $a$, $b$, $a^{-1}$, $b^{-1}$ (including empty string, denoted $e$). Examples: $ab$, $ba$, $a^{-1}b^{-1}ab^2a$, ....
- **Set of all words:** denoted $F$.
- **Spelling convention:** Simplify as much as possible. For example, $b^{-1}b^2$.



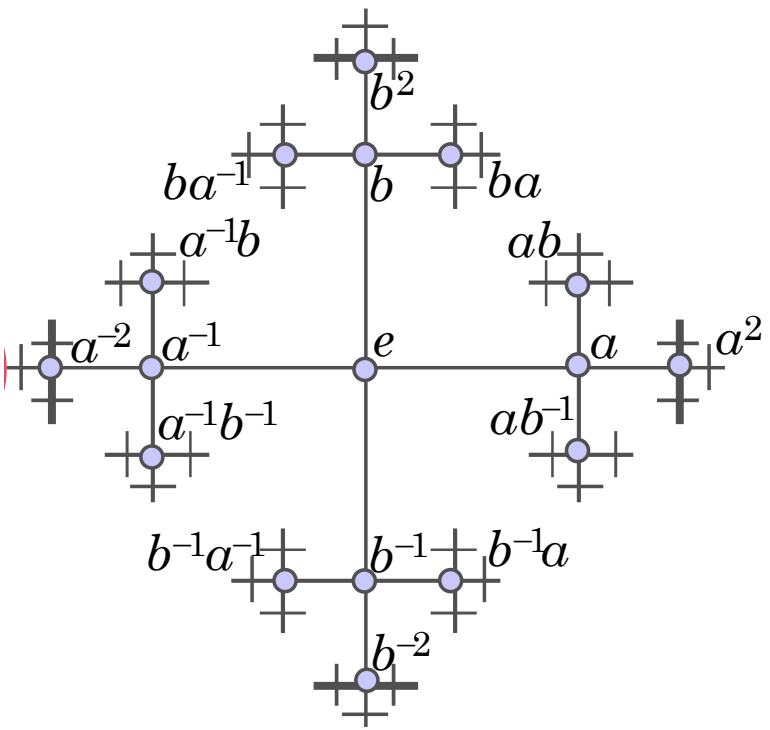
# Generalizing Scissors Congruence: Graphs and Alphabets

- **Words:** Finite strings of $a$, $b$, $a^{-1}$, $b^{-1}$ (including empty string, denoted $e$). Examples: $ab$, $ba$, $a^{-1}b^{-1}ab^2a$, ....
- **Set of all words:** denoted $F$.
- **Spelling convention:** Simplify as much as possible. For example, ~~$b^{-1}b^2$~~ $b$.



# Generalizing Scissors Congruence: Graphs and Alphabets

- **Words:** Finite strings of $a$, $b$, $a^{-1}$, $b^{-1}$ (including empty string, denoted $e$). Examples: $ab$, $ba$, $a^{-1}b^{-1}ab^2a$, ....
- **Set of all words:** denoted $F$.
- **Spelling convention:** Simplify as much as possible. For example, ~~$b^{-1}b^2$~~ $b$.



# Cutting and Gluing in $F$: Equidecomposability

Let $A$ = set of all words beginning with $a$ or $a^{-1}$. Then

$$A = \underbrace{A^+}\cup\underbrace{A^-}.$$

When working with subsets of $F$:

- **Cutting:** Decomposing into smaller subsets of $F$
- **Rearranging:** Multiplying by a letter (or word) on the left.
- **Equidecomposability ($A \sim B$):** $A, B \subset F$ are equidecomposable if we can cut $A$ into a finite number of smaller sets and rearrange the pieces to form $B$.

# Cutting and Gluing in $F$: Equidecomposability

Let $A$ = set of all words beginning with $a$ or $a^{-1}$. Then

$$A = \underbrace{A^+}_{\substack{\text{words}\\ \text{beginning}\\ \text{with } a}} \cup \underbrace{A^-}_{}.$$

When working with subsets of $F$:

- **Cutting:** Decomposing into smaller subsets of $F$
- **Rearranging:** Multiplying by a letter (or word) on the left.
- **Equidecomposability ($A \sim B$):** $A, B \subset F$ are equidecomposable if we can cut $A$ into a finite number of smaller sets and rearrange the pieces to form $B$.

# Cutting and Gluing in $F$: Equidecomposability

Let $A$ = set of all words beginning with $a$ or $a^{-1}$. Then

$$A = \underbrace{A^+}_{\substack{\text{words}\\ \text{beginning}\\ \text{with } a}} \cup \underbrace{A^-}_{\substack{\text{words}\\ \text{beginning}\\ \text{with } a^{-1}}} .$$

When working with subsets of $F$:

- **Cutting:** Decomposing into smaller subsets of $F$
- **Rearranging:** Multiplying by a letter (or word) on the left.
- **Equidecomposability ($A \sim B$):** $A, B \subset F$ are equidecomposable if we can cut $A$ into a finite number of smaller sets and rearrange the pieces to form $B$.

# Cutting and Gluing in $F$: Equidecomposability

Let $A$ = set of all words beginning with $a$ or $a^{-1}$. Then

$$A = \underbrace{A^+}_{\substack{\text{words}\\ \text{beginning}\\ \text{with } a}} \cup \underbrace{A^-}_{\substack{\text{words}\\ \text{beginning}\\ \text{with } a^{-1}}} .$$

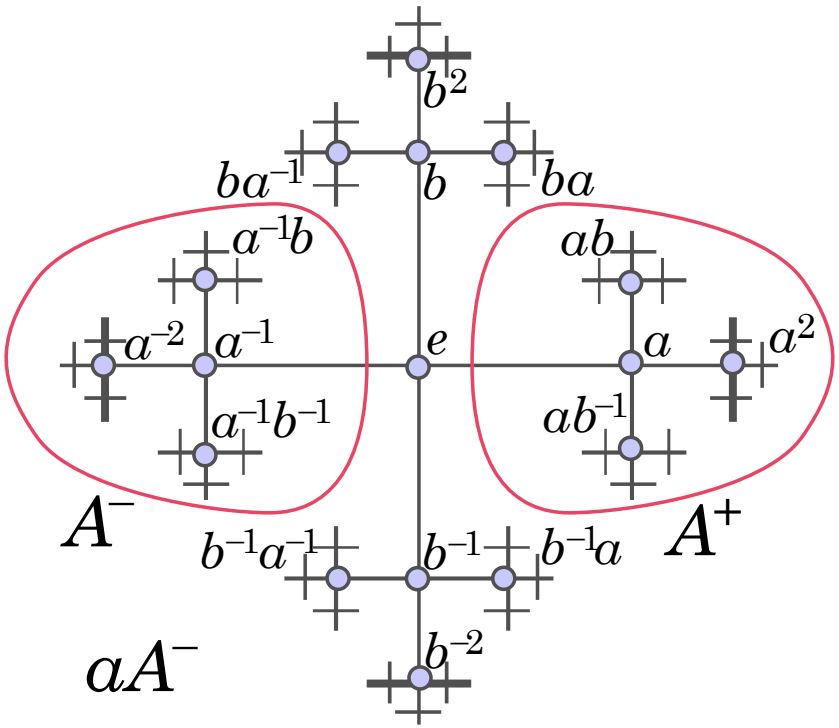
When working with subsets of $F$:

- **Cutting:** Decomposing into smaller subsets of $F$
- **Rearranging:** Multiplying by a letter (or word) on the left.
- **Equidecomposability ($A \sim B$):** $A, B \subset F$ are equidecomposable if we can cut $A$ into a finite number of smaller sets and rearrange the pieces to form $B$.

# Cutting and Gluing in $F$: Equidecomposability

Let $A$ = set of all words beginning with $a$ or $a^{-1}$. Then

$$A = A^+ \cup A^-.$$

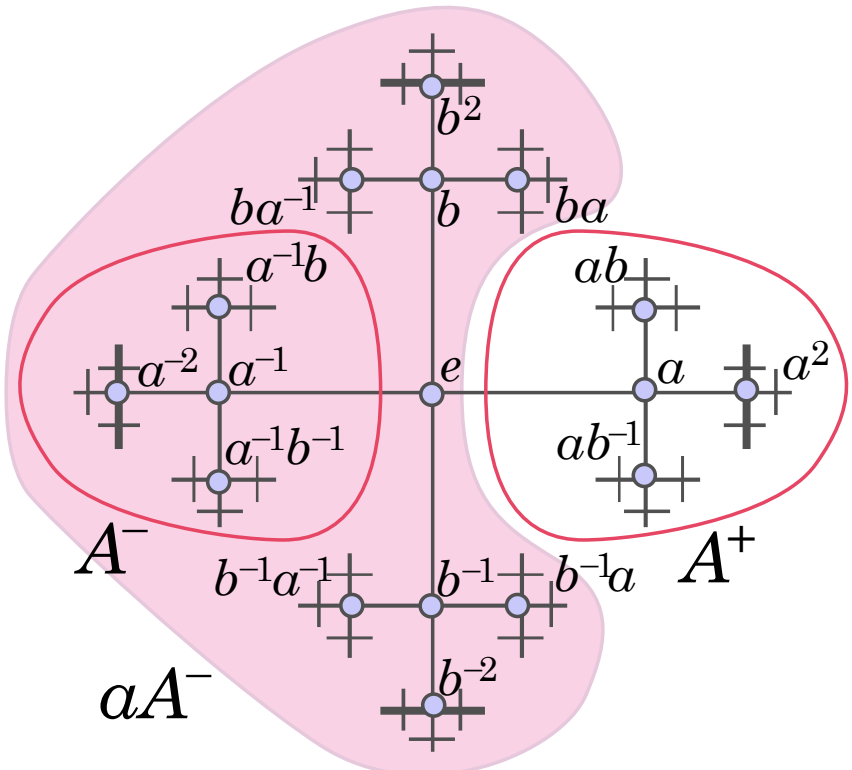



**In fact,** $F \sim$ (2 copies of $F$) !

# Cutting and Gluing in $F$: Equidecomposability

Let $A$ = set of all words beginning with $a$ or $a^{-1}$. Then

$$A = A^+ \cup A^-.$$



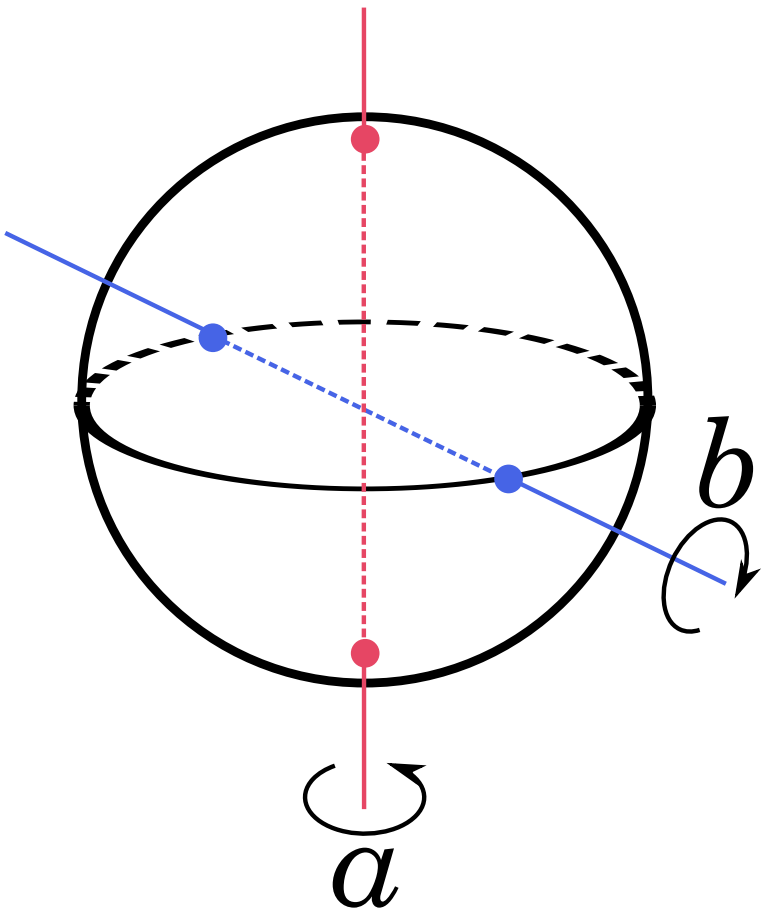

$aA^- \sim A^-$, and
$aA^- = B \cup A^- \cup \{e\}$

$$\Longrightarrow \quad A \sim A^+ \cup (B \cup A^- \cup \{e\})$$
$$A \sim F$$

$F$ is equidecomposable with a . . .
*subset of itself* !

**In fact,** $F \sim$ (2 copies of $F$) !

# Penultimate demonstration



Different notions of what you can choose to cut and glue can give rise to radically different notions of equidecomposability.

Thank you!

Yvonne Lai ○ yxl@umich.edu ○ U Michigan